Tejas Phutane

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SUMMARY

- Senior Robotics and Computer Vision engineer with **4+ years** of experience in robotics software, computer vision, motion planning and hardware integration. Currently contributing to humanoid robotics at FEV India, with hands-on work in simulation-first development, demonstration-data collection and ACT/SmolVLA training pipelines.
- Strong background in system architecture, sim-to-real deployment, perception, manipulation, digital twins and full-stack robotics integration across production-grade and experimental platforms. Developing capability in LLM applications, agentic AI and conversational robotics for Physical AI systems.

SKILLS

- **Robotics & Control:** Humanoid Robotics, Navigation Stack, Arm Manipulation, Motion Planning, Path Planning, Trajectory Optimization & Control, Kinematic Analysis, Localization and Sensor Fusion, Locomotion Policy Exploration.
- **Simulation & Frameworks:** ROS2, MoveIt, Gazebo, Isaac Sim,Mujoco, Sim-to-Real Workflows, Digital Twin Generation, Hardware-Abstraction Principles and Safety-Aware Development.
- **Computer Vision & AI:** OpenCV, YOLO (v3-v11), NVIDIA DeepStream, TensorRT, Vision-Language-Action (VLA) Models, Manipulation Policy Workflows,Behavioral Cloning, ACT and SmolVLA Training Pipelines.
- **AI & LLM Engineering:** LLM Application Development, Prompt Engineering, RAG Concepts, Agentic AI Workflows, OpenAI/Anthropic/Gemini APIs, Hugging Face and LangChain; Conversational Robotics and Embodied-AI Applications.
- **Programming Systems & DevOps:** C++ (strong), Python (advanced), C, MATLAB, Linux/UNIX, Git, Docker, AWS
- **Hardware & Edge Devices:** Unitree G1, SO-101 Arm Platform, NVIDIA Orin, Industrial Cameras, LiDAR, Multi-Axis Motion Controllers

PROFESSIONAL EXPERIENCE

FEV India - Pune, India ***March 2026 - Present***

- Senior Robotics Engineer L1

• Major deliveries and Contributions:

- Contributing to humanoid robotics development across simulation, robot-software integration, validation and real-world deployment workflows.
- Building **simulation environments and digital twins** in NVIDIA Isaac Sim, supporting motion and manipulation experimentation and sim-to-real workflows.
- Integrating and troubleshooting robotic hardware and software; applying modular software and **hardware-abstraction** principles to ROS 2-based integration.
- Applying **safety-aware development and hazard-analysis concepts** to support robust robot operation around people.
- Building **demonstration-data collection workflows** using egocentric, simulated and mimic-generated sources for robot-learning tasks.
- Working on behavioral-cloning workflows and ACT/SmolVLA training pipelines, developing hands-on capability in imitation-learning-based manipulation.

Wastefull Insights (Robotics) – Vadodara, India

- Senior Robotics Engineer ***Aug 2023 – Nov 2025***

- ***Major deliveries and Contributions:***
  - Led **software architecture**, **perception pipeline** development, and technical team coordination for autonomous robotic systems. Managed end-to-end project delivery from design through production deployment.
  - Architected **motion planning** and **control algorithms** for **3-axis gantry systems**, reducing inertial motion losses by 15% and improving system efficiency by 80%. Increased overall system speed and accuracy by 40% through **iterative algorithm refinement** and **trajectory optimization.**
  - Coordinated with hardware vendors for camera, lighting, and motor integration; ensured seamless hardware-software synchronization.
  - Built **robot-vision analysis framework** with database integration for continuous performance monitoring and optimization. Developed regression models for throughput prediction with 91% accuracy.
  - Engineered custom multi-robot coordination algorithms for waste sorting facility processing 500+ objects/hour; optimized task allocation reducing idle time by 30% and increasing overall throughput by 25%.

- Robotics Engineer ***Aug 2022 –Sep 2023***

- Refactored codebase from Python to C++ for performance-critical modules, reducing computational overhead by 80% and enabling real-time processing.
- Established 2x faster and memory optimized multimodel YOLO pipeline for detection and accurate tracking of waste objects using Nvidia Deep stream framework.

Engineering Services International (Science City, Robotics gallery) – Ahmedabad, India ***Oct 2021 – Aug 2022***

- ***Junior Robotics Engineer***
  - Architected perception and navigation stacks for autonomous mobile robot platforms using ROS ecosystem. Designed and validated motion planning algorithms in Gazebo simulation environment, reducing development cycle by **60%**.
  - Implemented **multi-modal sensor fusion** combining camera and **Lidar data with Kalman filtering** for robust localization.
  - Developed object detection pipeline using **OpenCV** and **YOLOv3** for real-world robot vision applications.
  - Developed and implemented fast and robust algorithms for Disc detection and trajectory estimation using OpenCV and ROS. Worked on **Epson Scara robot** and implemented defense and attack strategies based on trajectory and position of disc.

IN2PETA Services – Hyderabad, India ***June 2021 – Oct 2021***

- ***Computer Vision and ROS Intern***
  - Developed an **autonomous mobile pick-and-place robot** for cluttered toy environments, integrating perception, planning, and control using the **ROS framework**.
  - Implemented and tested **object detection, high-level task planning, path planning, and sensor fusion** modules on **NVIDIA Jetson** for real-time robotic operation.
  - Built **computer vision pipelines using OpenCV** for object detection and scene understanding in dynamic environments.

PROJECTS

Precision Manufacturing Vision QC System **Oct 2025**

- Conducted reverse engineering of legacy control systems and PLC programming Developed high-performance defect detection model trained on 5000+ annotated samples.
- Integrated hardware stack including industrial cameras, line-scan optics, and precision lighting Coordinated hardware procurement and mechanical system calibration with vendors.
- Reduced manual inspection labor by 60% and improved defect detection consistency and established continuous deployment pipeline for model updates and performance validation.

Multi-Stream Vision Analytics Platform **Oct 2025**

- **Technologies:** Python, C++, YOLOv5, NVIDIA Deep Stream, TensorRT, Kafka, Timescale DB, PostgreSQL, Docker, Streamlit, REST APIs
- Real-time video processing system analyzing multiple RTSP feeds simultaneously.
- YOLOv5-based object detection with secondary classification for SKU identification and counting
- End-to-end pipeline: NVIDIA Deep Stream ingestion, TensorRT inference optimization, Kafka message streaming, Timescale DB time-series storage
- Achieved 99%+ detection accuracy with sub-50ms latency per frame
- Integrated REST APIs for real-time analytics queries and dashboard visualization using Streamlit.
- Deployed to production environment with continuous monitoring and model retraining pipeline.

Waste Sorting Robot-Vision Analysis Framework **Aug 2025**

- Built **database-backed analytics framework** to log robot operations and vision detections across production runs.
- Implemented **performance analysis and trend extraction tools**, enabling visibility into throughput, cycle time, and system bottlenecks.
- Developed **regression models** to predict system throughput based on operational parameters and robot state data.
- Created **real-time and historical visualization dashboards** for monitoring system health and performance.
- Enabled **data-driven optimization** of robot workflows, reducing overall cycle time by **15%**

Autonomous Warehouse Management System (Eyantra Robotics Competition) | LINK **Sep 2020**

- Developed autonomous warehouse management system using two UR5 robots in ROS Gazebo environment ,used Python nodes for controlling robot arms using MoveIt motion planning framework.
- Implemented IOT using MQTT protocol for communication between robots and visualized package delivery progress on google spreadsheets using Google App Scripting.

CERTIFICATIONS

- **Introduction to Robotics** — NPTEL
- **Robotics Specialization** — Coursera
- **Control of Mobile Robots** — Coursera

EDUCATION

- **University of Mumbai, MCT's Rajiv Gandhi Institute of Technology** **Mumbai, India**
  Bachelor of Engineering, Electronics & Telecommunication Engineering (CGPA: 7.04) *March 2021*